C4.1 About an initiating master

This section describes the behavior of an initiating master. Typically, an initiating master issues a transaction to progress an internal action such as a load or store operation.

The internal action requires:

- for a load, the master must get the data from either:
  - a valid copy of the appropriate cache line
  - a transaction that returns valid read data.
- for a store, the master needs permission to store the cache line from either:
  - a copy of the appropriate cache line in a Unique state
  - a transaction type that gives the master permission to store the cache line.

C4.1.1 Transaction groups

The following sections describe the expected channel activity for the transaction groups:

- *Read transactions*
- *Clean transactions*
- *Make transactions* on page C4-185
- *Write transactions* on page C4-185
- *Evict transactions* on page C4-185.

Read transactions

The read transaction group is:

- ReadNoSnoop
- ReadOnce
- ReadClean
- ReadNotSharedDirty
- ReadShared
- ReadUnique.

A Read transaction progresses as follows:

1. The address is issued on the read address (AR) channel.
2. The data and response is returned on the read data (R) channel. The number of data beats required is determined by **ARLEN**.
3. Completion of a Read transaction is signaled by the master asserting **RACK**.

Clean transactions

The clean transaction group is:

- CleanUnique
- CleanShared
- CleanInvalid.

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A Clean transaction progresses as follows:

1. The address is issued on the AR channel.
2. A single transfer on the R channel returns the response. No data is returned for a Clean transaction.
3. Completion of a Clean transaction is signaled by the master asserting **RACK**.

Make transactions

The make transaction group is:

- MakeUnique
- MakeInvalid.

For the initiating master, a Make transaction progresses as follows:

1. The address is issued on the AR channel.
2. A single transfer on the R channel returns the response. No data is returned for a Make transaction.
3. Completion of a Make transaction is signaled by the master asserting **RACK**.

Write transactions

The write transaction group is:

- WriteNoSnoop
- WriteUnique
- WriteLineUnique
- WriteBack
- WriteClean.

For the initiating master, a Write transaction progresses as follows:

1. The address is issued on the AW channel.
2. The data is transferred on the W channel.
3. The response is returned on the B channel.
4. Completion of a Write transaction is signaled by the master asserting **WACK**.

Evict transactions

The evict transaction group is, Evict.

For the initiating master, an Evict transaction progresses as follows:

1. The address is issued on the AW channel.
2. The response is returned on the B channel. No data is transferred for an Evict transaction.
3. Completion of an Evict transaction is signaled by the master asserting **WACK**.

Read barrier transactions

For the master initiating the transaction, a Read Barrier transaction progresses as follows:

1. The transaction is issued on the AR channel.
2. A single transfer on the R channel returns the response. No data is returned for a Read Barrier transaction.
3. Completion of a Read Barrier transaction is signaled by the master asserting **RACK**.

See Chapter C8 *Barrier Transactions*.

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Write barrier transactions

For the master initiating the transaction, a Write Barrier transaction progresses as follows:

1. The transaction is issued on the AW channel.
2. The response is returned on the B channel. No data is transferred for a Write Barrier transaction.
3. Completion of a Write Barrier transaction is signaled by the master asserting **WACK**.

See Chapter C8 *Barrier Transactions*.

DVM transactions

For the master initiating the transaction, a DVM transaction progresses as follows:

1. The transaction is issued on the AR channel.
2. A single transfer on the R channel returns the response. No data is returned for a DVM transaction.
3. Completion of a DVM transaction is signaled by the master asserting **RACK**.

See Chapter C12 *Distributed Virtual Memory Transactions*.

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